

Characterization of Flavone and Flavonol Aglycones by Collision-induced Dissociation Tandem Mass Spectrometry†

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A mass spectrometric method based on the combined use of fast atom bombardment collisionally-induced dissociation (CID) and tandem mass spectrometry has been used for the structural characterization of free and conjugated flavone and flavonol aglycones. Low-energy CID spectra of the $[M + H]^+$ (or Y_0^+) ions show simple fragmentation patterns, which allow characterization of the substituents in the A and B rings and, in particular, differentiation between flavones and flavonols. A systematic nomenclature for product ions produced from protonated molecules under CID conditions is proposed. © 1997 by John Wiley & Sons, Ltd.

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Flavones and flavonols are the two largest groups of naturally occurring flavonoids in plants. They are mainly present as *O*- and/or *C*-glucosides, but free aglycones have also been found in various plants.^{1,2} Both aglycones and their glycosides show a broad range of biological activities and are of interest for chemotaxonomy. Application of modern mass spectrometric methods to quickly identify and characterize these polyphenolic secondary plant metabolites is therefore particularly relevant.

Mass spectra obtained under electron impact (EI) conditions have been widely used for structural investigation of flavonoids.^{3–6} This ionization technique has been applied to all classes of flavonoid aglycones. Structurally informative fragmentation, such as retro Diels–Alder reactions (RDA), allow one to infer the oxygenation or substitution pattern in the A and B rings. However, complex fragmentations may occur during EI due to the broad spread of internal energy carried by the initially produced $M^{+\cdot}$ ions,⁷ which may hide or suppress the peak for $M^{+\cdot}$ ions and for some important primary fragment ions containing structural information. Furthermore, EI shows limitations for the analysis of polyphenolic flavonoids, such as flavonols and *O*-glycosides, because of their low volatility and high polarity. Following the introduction of fast atom bombardment (FAB)⁸ as a standard ionization technique, useful structural information on polar biomolecules could be obtained directly and easily, especially by combining FAB with collisionally induced dissociation (CID) and tandem mass spectrometry (MS/MS). Consequently, the latter methodology was successfully applied to the structural characterization of flavones,⁹ flavonols^{9,10} and their *O*- and *C*-glycosides.^{10–15}

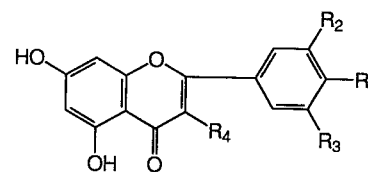
In the present study we have examined in detail the low-energy CID behaviour of the protonated forms of naturally occurring flavones and flavonols. Furthermore, we have also evaluated whether the aglycone part in *O*-glycosides can be characterized, by subjecting the Y_0^+ ion, which corresponds to the protonated

aglycone and is generated upon FAB, to low-energy CID. This method has advantages for the phytochemical analysis of flavonoid-*O*-glycosides because the chemical hydrolysis and subsequent purification procedures which are necessary for the identification of the aglycone part can be circumvented.

EXPERIMENTAL

Materials

The flavone and flavonol aglycones (1–8) investigated in this study are listed in Fig. 1. Apigenin (1), luteolin (2), acacetin (3), chrysoeriol (4), isorhamnetin (8) and two flavonoid glycosides, luteolin-7-*O*-glucoside (9) and kaempferol-7-*O*-neohesperidoside (10) (Fig. 2), were purchased from Roth (Karlsruhe, Germany). Kaempferol (5), quercetin (6) and myricetin (7) were isolated from plants. The purity of all compounds was checked by reversed-phase high-performance liquid chromatography (RPHPLC) on a ChromSpher 5 C₁₈ column (Chrompack, Middelburg, The Netherlands).



	trivial name	$[M+H]^+$	R ₁	R ₂	R ₃	R ₄
1	apigenin	271	OH	H	H	H
2	luteolin	287	OH	OH	H	H
3	acacetin	285	OCH ₃	H	H	H
4	chrysoeriol	301	OH	OCH ₃	H	H
5	kaempferol	287	OH	H	H	OH
6	quercetin	303	OH	OH	H	OH
7	myricetin	319	OH	OH	OH	OH
8	isorhamnetin	317	OH	OCH ₃	H	OH

Figure 1. Structures of the flavones (1–4) and flavonols (5–8) investigated.

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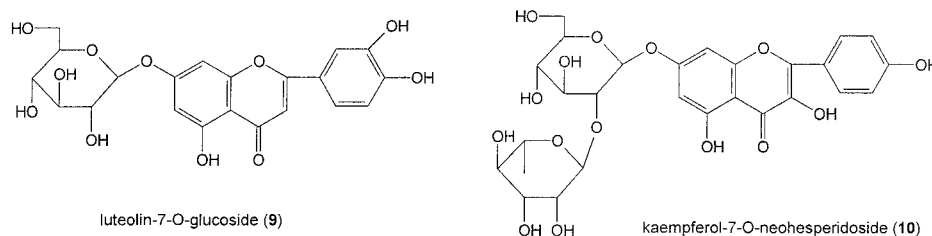


Figure 2. Structures of the flavonoid glycosides (**9**, **10**) investigated.

Mass spectrometry

All spectra were obtained on a VG 70-SEQ (Micro-mass, Manchester, UK) hybrid mass spectrometer with EBQQ configuration, equipped with a cesium ion source. Cesium ions with an impact energy of approximately 22 keV and a beam flux of 0.3 μ A were used as the ionization beam. The acronym FAB is used throughout the paper to refer to cesium ion bombardment. The acceleration voltage in the source was 8 kV. The samples were dissolved in dimethylsulphoxide (DMSO) or methanol (10 μ g/ μ L) and 1 μ L of the solution was mixed with 2 μ L of the liquid matrix (glycerol) on the stainless-steel probe tip.

For recording low-energy CID spectra, precursor ions were mass-selected with a resolution of about 1000 using the magnetic sector. CID was performed in the quadrupole gas cell with argon as collision gas at a pressure of approximately 5×10^{-6} mBar (measured in the quadrupole housing) and using a collision energy of 30 eV except if mentioned otherwise. The product-ion spectra were obtained in the continuous mode of acquisition with accumulation of fifteen scans and using unit resolution of the quadrupole analyser.

Nomenclature

The widely used nomenclature system for flavonoid aglycones, developed by Mabry and Markham,³ has been applied to define the various A and B ring fragments generated by EI. In the course of our mass spectral studies on flavonoid glycosides, we have found it inadequate for describing the range of product ions formed when protonated molecules are subjected to CID.^{9,10} A more systematic ion nomenclature for flavonoid aglycones is proposed that is conceptually similar to that introduced for the description of carbohydrate fragmentations in CID spectra of glycoconjugates.¹⁶ For free aglycones, the $^{i,j}\mathbf{A}^+$ and $^{i,j}\mathbf{B}^+$ labels are used to designate primary product ions containing intact A and B rings, respectively, in which

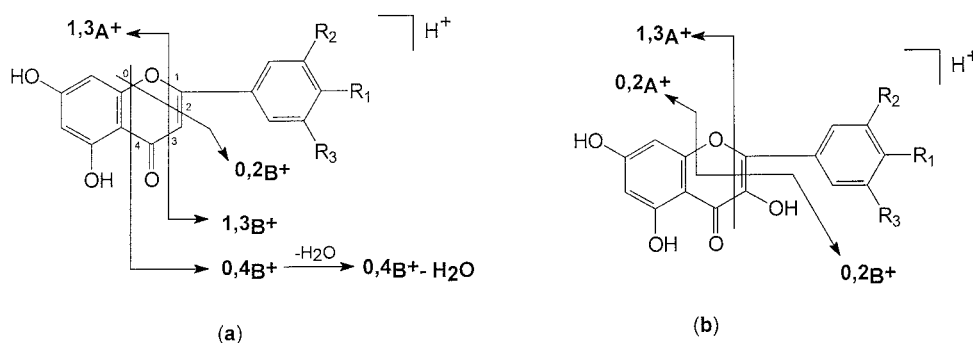
the superscripts *i* and *j* indicate the C-ring bonds that have been broken (see Scheme 1). For conjugated aglycones, an additional subscript 0 to the right of the letter is used to avoid confusion with the $\mathbf{A}_i^+$ and/or $\mathbf{B}_i^+$ ($i \geq 1$) labels that have been used to designate carbohydrate fragments containing a terminal (non-reducing) sugar unit.¹⁶ The $^{i,j}\mathbf{A}^+$ and $^{i,j}\mathbf{B}^+$ ions can undergo further fragmentation by losing small neutral molecules, such as H_2O and CO or combinations thereof. These product ions are denoted by the combined use of $^{i,j}\mathbf{A}^+$ (or $^{i,j}\mathbf{B}^+$) and the lost molecules, e.g. the $^{0,4}\mathbf{B}^+ - \text{H}_2\text{O}$ ion. For simplicity, the ions formed by the direct loss of radicals or small neutral molecules such as a methyl group, water, carbon monoxide and ketene from the $[\text{M} + \text{H}]^+$ ion, are not indicated with special labels.

RESULTS AND DISCUSSION

Fragmentation of flavone aglycones

The low-energy CID spectra of $[\text{M} + \text{H}]^+$ ions of **1–4** are summarized in Table 1; the spectrum and main fragmentations for **2** are illustrated in Fig. 3 and Scheme 2 respectively. The CID pathways observed for these four compounds (**1–4**) exhibit some common features but also show interesting differences. The most useful fragmentations in terms of flavone aglycone identification are those that involve cleavage of two C–C bonds at positions 1/3, 0/2 and 0/4 of the C ring, resulting in structurally informative $^{i,j}\mathbf{A}^+$ and $^{i,j}\mathbf{B}^+$ ions. These three fragmentation routes are designated below as pathway I, II and III respectively (Scheme 2), of which pathways I and III correspond to retro Diels–Alder fragmentations.

Pathway I gives rise to two species, i.e. a $^{1,3}\mathbf{A}^+$ ion at m/z 153 in the case of **1**, **2**, **3** and **4**, and a $^{1,3}\mathbf{B}^+$ ion at m/z 119, 135, 133 and 149 for **1**, **2**, **3** and **4** respectively. This pair of product ions clearly provides the substitution pattern in the A and B rings of each compound. Possible mechanisms for the formation of the $^{1,3}\mathbf{A}^+$ and



Scheme 1. Nomenclature and diagnostic product ions of protonated (a) flavones and (b) flavonols formed under low-energy CID.

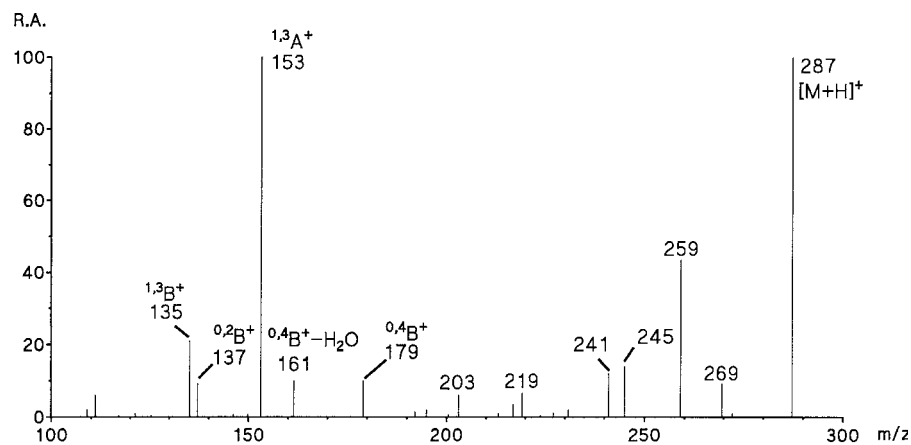
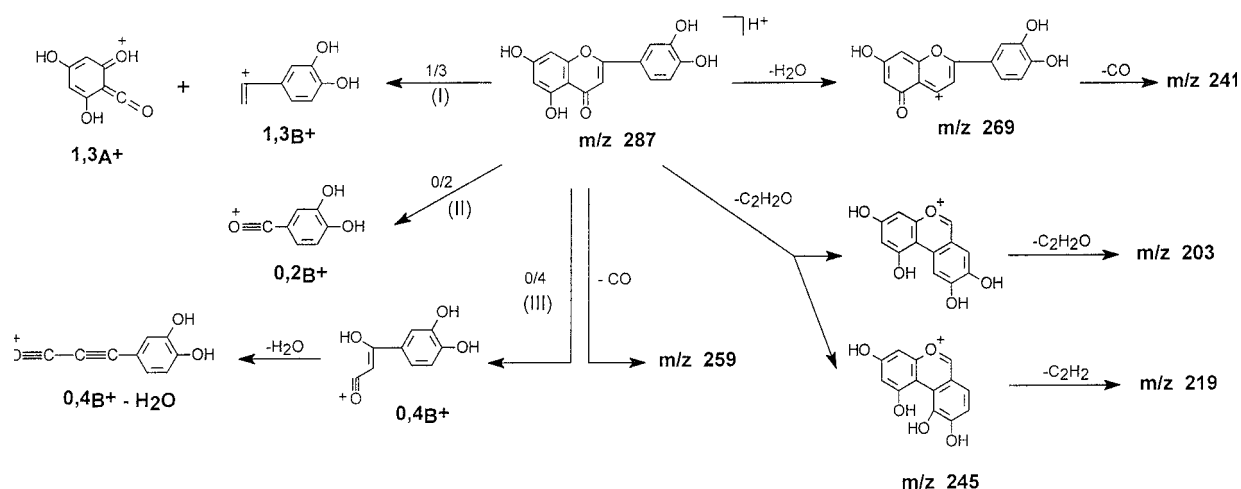


Figure 3. Low-energy CID spectrum obtained for protonated luteolin.

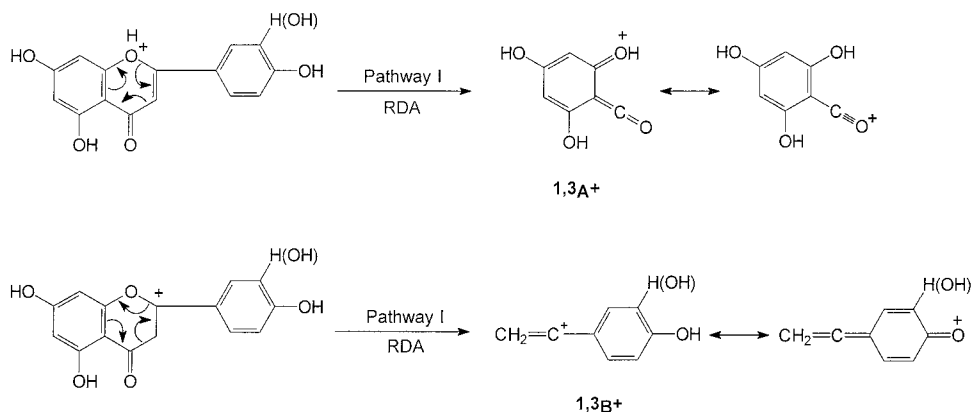


Scheme 2. Main fragmentations of protonated luteolin by low-energy CID.

$^{1,3}\text{B}^+$ ions in the case of **1** and **2** are proposed in Scheme 3. Protonation at two positions, the ether oxygen atom and the C-3 position in the C ring, may lead to two different protonated molecule species. In each of them an RDA fragmentation may occur, giving rise to $^{1,3}\text{A}^+$ or $^{1,3}\text{B}^+$ ions. Analogy exists with the mechanism of the RDA reaction under EI conditions, that is assumed to be charge-driven.^{17,18} The difference between RDA reactions observed under EI and FAB CID conditions can easily be rationalized; in the former case, the RDA reaction is induced by a radical site and the fragment

ions formed are odd-electron ions, while in the latter, it is initiated by the even-electron charge and even-electron ions are formed as direct products. The $^{1,3}\text{A}^+$ ions are the base peaks for **1** and **2** and correspond to relatively abundant ions for **3** and **4** (Table 1). A driving force for these RDA processes may be the formation of ions that are stabilized by resonance.

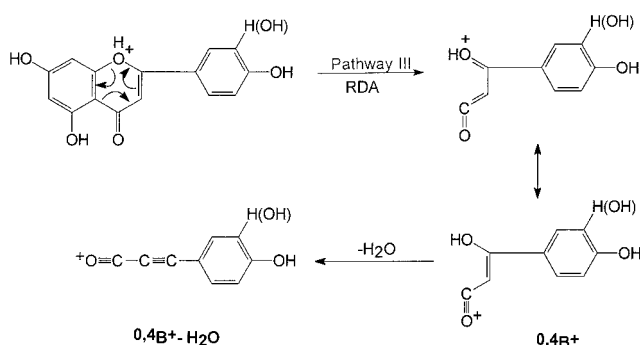
Pathway III, an alternative RDA fragmentation, results in $^{0,4}\text{B}^+$ ions, that can further fragment by loss of a molecule of water, giving rise to $^{0,4}\text{B}^+ - \text{H}_2\text{O}$ ions (Scheme 4). The prominent ions at m/z 179 and 161 are



Scheme 3. Mechanisms proposed for the formation of $^{1,3}\text{A}^+$ and $^{1,3}\text{B}^+$ ions in the case of the protonated flavones **1** and **2**.

Table 1. Principal product ions formed from $[M+H]^+$ ions using low-energy CID for the flavone aglycones **1**, **2**, **3** and **4**

ions	1	2	3	4
$[M+H-CH_3]^+$	—	—	270 (100) ^a	286 (100)
$[M+H-H_2O]^+$	253 (7)	269 (9)	267 (0.4)	283 (4.4)
$[M+H-CO]^+$	243 (32)	259 (44)	257 (3.5)	273 (3.2)
$[M+H-CH_3OH]^+$	—	—	253 (0.4)	269 (1.4)
$[M+H-C_2H_5O]^+$	229 (21)	245 (14)	243 (3.6)	259 (1.6)
$[M+H-CH_3-CO]^+$	—	—	242 (1.3)	—
$[M+H-H_2O-CO]^+$	225 (12)	241 (12)	239 (1.9)	—
$[M+H-C_2H_5O-C_2H_5]^+$	203 (8)	219 (7)	217 (2.1)	—
$[M+H-C_2H_5O-C_2H_5O]^+$	—	203 (6)	—	—
$^{1,3}A^+$	153 (100)	153 (100)	153 (9.7)	153 (3.0)
$^{0,4}B^+$	163 (7)	179 (10)	177 (0.6)	—
$(^{0,4}B^+ - H_2O)$	145 (12)	161 (10)	159 (2.4)	175 (0.7)
$^{0,2}B^+$	121 (10)	137 (9)	135 (1.1)	151 (0.9)
$^{1,3}B^+$	119 (24)	135 (21)	133 (4.3)	149 (1.0)

^a m/z (% relative abundance)**Scheme 4.** Mechanisms proposed for the formation of $^{0,4}B^+$ and $(^{0,4}B^+ - H_2O)$ ions in the case of protonated **1** and **2**.

due to these cleavages in the case of **2**, while the ions at m/z 163 and 145, m/z 177 and 159, respectively, are formed by the same processes in the case of **1** and **3**. This indicates that III corresponds to a general fragmentation pathway for flavone aglycones. It is worth noting that this RDA fragmentation is not observed under either high-energy FAB CID⁹ or EI conditions.³ Furthermore, only protonated flavone aglycones appear to fragment through pathway III under low-energy FAB CID conditions, so that the $^{0,4}B^+$ and $(^{0,4}B^+ - H_2O)$ ions are diagnostic of flavone aglycones. Pathway II leads to the formation of $^{0,2}B^+$ ions at m/z 121 for **1**, m/z 137 for **2**, m/z 135 for **3** and m/z 151 for **4**, respectively. This fragmentation will be discussed later.

In addition to the $^{ij}A^+$ and $^{ij}B^+$ ions already discussed, there are losses of small molecules and/or radicals from the $[M+H]^+$ ion that are useful for identifying the presence of specific functional groups. For example, the presence of a methoxy group is easily detected by the loss of 15 u (CH_3) and 32 u (CH_3OH)

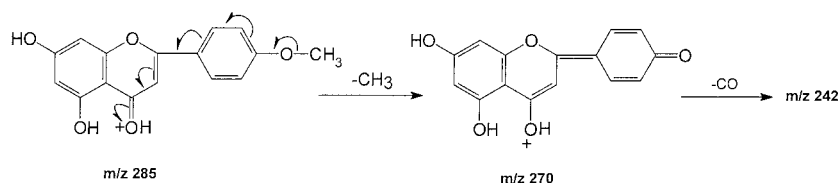
from the $[M+H]^+$ ion (Table 1). The former cleavage results in the base peaks at m/z 270 and 286 for **3** and **4** respectively. This rather uncommon transition from an even-electron to an odd-electron ion is found to be characteristic of a phenolic methyl ether group. In the case of **3**, (see Scheme 5) the further loss of 28 u (CO) gives rise to the $[M+H-CH_3-CO]^+$ ion (m/z 242). The driving force may be the formation of an odd-electron $[M-CH_3]^+$ ion with a highly delocalized charge.

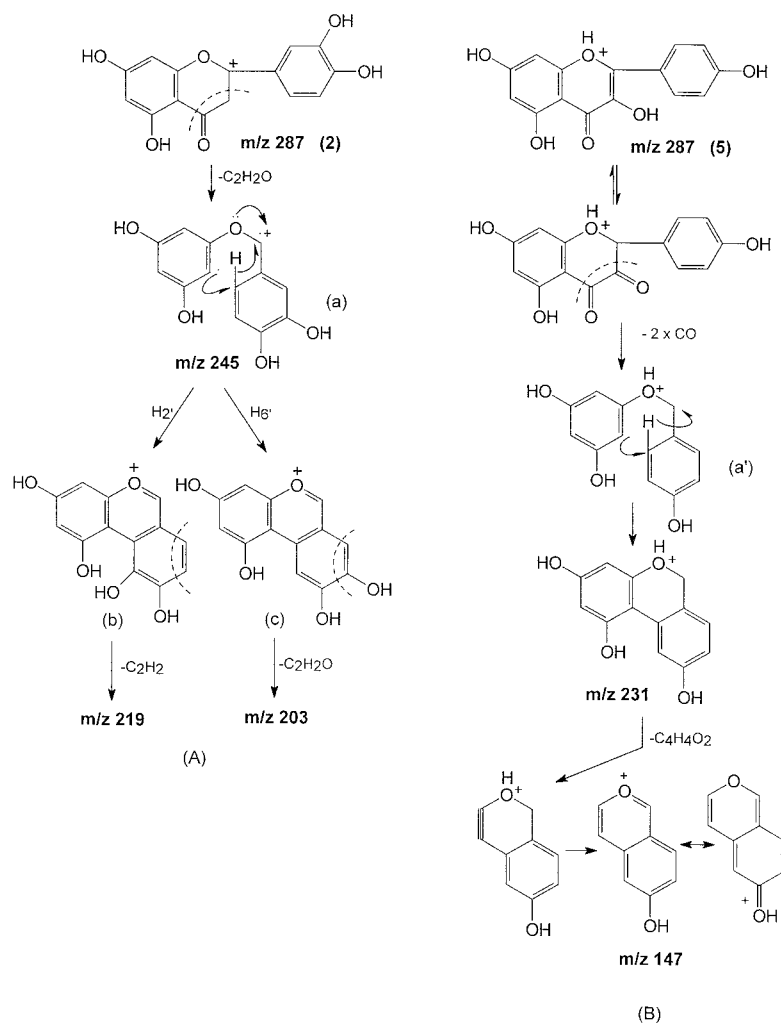
The losses of some small neutral molecules from the $[M+H]^+$ ion of luteolin **2**, as an example, are rationalized in Scheme 2. For **1** and **2**, the successive losses of 18 u (H_2O) and 28 u (CO) from $[M+H]^+$ ions give rise to the ions at m/z 253 and 225 for **1** and m/z 269 and 241 for **2**. The loss of H_2O may be induced by protonation at the 4-keto group of the C ring and has also been observed in high-energy CID by Baracco *et al.*⁹ The loss of 28 u (CO) and 42 u (C_2H_5O) from $[M+H]^+$ ions is observed in the case of **1**, **2**, **3** and **4** (Table 1). The latter loss could be explained by protonation at C-3 and subsequent cleavage of bonds 2 and 4 in the C ring, resulting in an ion with an open structure (**a**) which could rearrange to a more stable one (**b** or **c**) (Scheme 6(a)). This mechanism is supported by the occurrence of additional fragments at m/z 203 for **1**, m/z 203 and 219 for **2**, and m/z 217 for **3** (Table 1). These data suggest that a hydrogen atom (H-6' or H-2') in the B ring is involved in the proposed rearrangement process that leads to structures **b** and **c**.

Fragmentation of flavonol aglycones

The low-energy CID spectra of $[M+H]^+$ ions of the flavonol aglycons **5–8**, are summarized in Table 2; the spectrum and main fragmentations for **5** are illustrated in Fig. 4 and Scheme 7. Compared with flavones, the CID spectra of the protonated flavonols **5**, **6**, **7** and **8** are more complex. Although pathway III is not observed in these spectra, the occurrence of a 3-OH group in the C-ring appears to result in more possibilities for fragmentation (Table 2). The two fragmentation routes, pathways I and II, give rise to $^{1,3}A^+$ and $^{0,2}A^+$ ions at m/z 153 and 165, respectively, for all four flavonols (**5–8**) and $^{0,2}B^+$ ions at m/z 121, 137, 153 and 151 for **5**, **6**, **7** and **8**, respectively. Pathway I results in the $^{1,3}A^+$ ion, but the $^{1,3}B^+$ ion that is formed in the case of all flavones (**1–4**) is not observed. Instead, a $(^{1,3}B^+ - 2H)$ ion is found at m/z 133, 149, 165 and 163 for **5**, **6**, **7** and **8**, respectively.

A plausible mechanism for the formation of $^{0,2}A^+$ and $^{0,2}B^+$ ions (pathway II) is proposed in Scheme 8. Protonation at the C-3 and C-2 positions followed by cleavage of bonds 0 and 2 in the C ring may explain the formation of $^{0,2}A^+$ and $^{0,2}B^+$ ions. A protonated 1,2-diketobutene ring for the $^{0,2}A^+$ ion was proposed

**Scheme 5.** Mechanisms proposed for the formation of the $[M+H-CH_3]^+$ and $[M+H-CH_3-CO]^+$ ions in the case of **3**.



Scheme 6. Mechanisms proposed for the formation of (A) $[M + H - C_2H_2O]^+$ ions for **2** and (B) $[M + H - C_2O_2]^+$ ions for **5** under low-energy CID conditions.

Table 2. Principal product ions formed from $[M+H]^+$ ions using low-energy CID for the flavonol aglycones **5**, **6**, **7** and **8**

ions	5	6	7	8
$[M+H-CH_3]^+$	—	—	—	302 (100) ^a
$[M+H-H_2O]^+$	269 (13)	285 (60)	301 (32)	299 (4.7)
$[M+H-CO]^+$	259 (85)	275 (100)	291 (76)	289 (9.5)
$[M+H-CH_3OH]^+$	—	—	—	285 (31)
$[M+H-C_2H_2O]^+$	245 (19)	261 (9)	277 (7)	275 (3.2)
$[M+H-H_2O-CO]^+$	241 (37)	257 (74)	273 (100)	271 (3.2)
$[M+H-2CO]^+$	231 (30)	247 (21)	263 (19)	261 (3.4)
$[M+H-CH_3OH-CO]^+$	—	—	—	257 (5.7)
$[M+H-H_2O-2CO]^+$	213 (39)	229 (60)	245 (41)	—
$^{0,2}A^+$	165 (90)	165 (73)	165 (81)	165 (6.8)
$^{1,3}A^+$	153 (100)	153 (39)	153 (63)	153 (4.4)
$[M+H-2CO-C_4H_4O_2]^+$	147 (19)	163 (14)	179 (12)	177 (1.7)
$(^{0,2}A^+-CO)$	137 (20)	137 (48)	137 (12)	137 (1.4)
$(^{1,3}B^+-2H)$	133 (24)	149 (21)	165 (81)	163 (2.2)
$(^{1,4}A^++2H)$	127 (14)	127 (11)	127 (11)	127 (1.4)
$^{0,2}B^+$	121 (81)	137 (48)	153 (39)	151 (5.3)
$(^{1,3}A^+-C_2H_2O)$	111 (22)	111 (16)	111 (20)	111 (1.4)
other ions	—	121 (13)	137 (12)	139 (6.3)

^a m/z (% relative abundance)

earlier by de Koster *et al.*¹⁰, who also observed this ion in the first-order FAB spectrum. It is worth noting that the $^{0,2}A^+$ ion is present with a relatively high abundance in the metastable-ion spectra of flavonols,¹⁰ but is hardly observed under high-energy CID conditions⁹ suggesting that the formation of the $^{0,2}A^+$ ion corresponds to a low-energy fragmentation process. In order to support this hypothesis, CID spectra of protonated kaempferol were recorded at three different collision energies, i.e. 10, 30 and 50 eV. The relationship between relative abundance and collisional energy for the $^{1,3}A^+$, $^{0,2}A^+$ and $^{0,2}B^+$ ions is shown in Fig. 5. It is clear that their relative abundances depend on the collisional energy. The $^{0,2}A^+$ ion decreases, while the $^{0,2}B^+$ ion increases, upon increasing the collision energy, indicating that the $^{0,2}A^+$ ion is favoured at lower energy (≤ 30 eV) while the reverse holds for the $^{0,2}B^+$ ion. It is worth mentioning that the $^{1,3}A^+$, $^{0,2}A^+$ and $^{0,2}B^+$ ions all appear as strong peaks at 30 eV, with comparable relative abundances, a pattern which can be regarded as a characteristic feature of flavonol aglycones.

As was found for fragmentation of protonated flavones, ions corresponding to the loss of H_2O , CO , ketene and their combined loss from $[M + H]^+$ ions are also observed in the high-mass region. However,

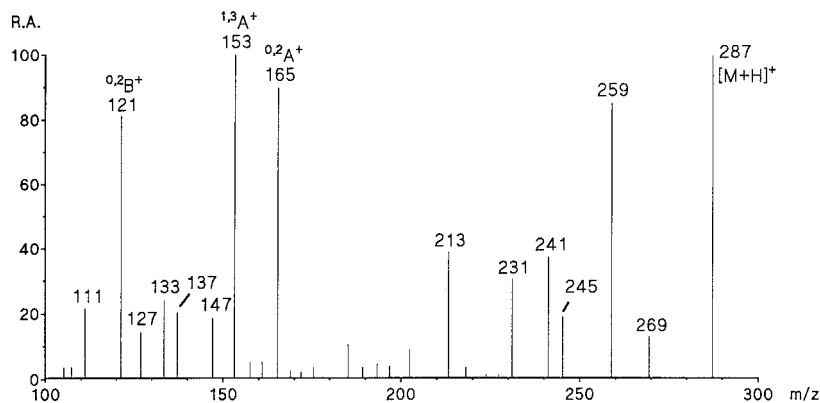
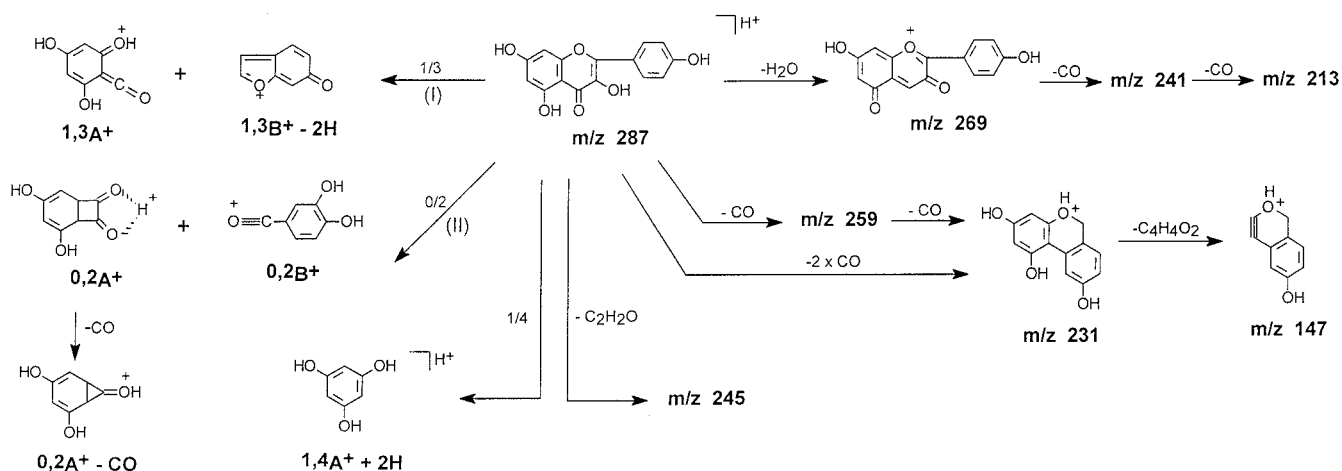
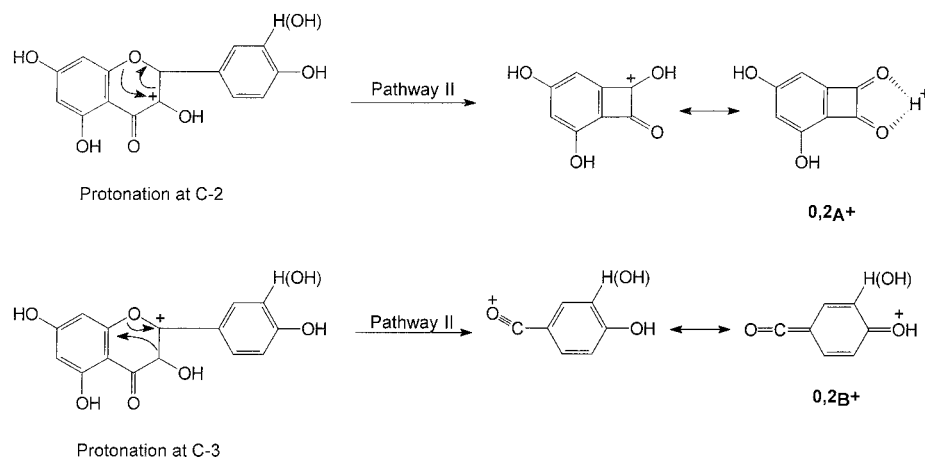


Figure 4. Low-energy CID spectrum obtained for protonated kaempferol.



Scheme 7. Main fragmentations of protonated kaempferol at low-energy CID.



Scheme 8. Mechanisms proposed for the formation of $^{0,2}A^+$ and $^{0,2}B^+$ ions in the case of protonated **5** and **6**.

their ion structures may be different from those proposed in the case of flavones. For **5**, for example, the ions at m/z 269, 241 and 213 could be explained by the successive losses of a molecule of H_2O and two molecules of CO .⁹ The ions at m/z 259 and 231 correspond to the successive losses of two CO molecules. The latter ion could also be formed directly from the $[M + H]^+$ ion by cleavage of bonds 2 and 4, involving an ion with an open structure (a') and H-6' (or H-2') rearrangement (Scheme 6(b)). The further cleavage of the A ring gives rise to the m/z 147 ion.

Due to the presence of a methoxy group in the B

ring, a fragmentation pattern different from that of **5**, **6** and **7** is found for isorhamnetin. Four major product ions at m/z 165, 153, 151 and 139 can be noted in the middle mass range (100–200 u). The former three are assigned to $^{0,2}A^+$, $^{1,3}A^+$ and $^{0,2}B^+$, respectively, indicating that **8** also follows the same fragmentation routes as those found for other flavonols containing free hydroxyl groups. The ion at m/z 139, however, could not be explained by a straightforward mechanism. In contrast to the fragmentation observed for flavonols containing free hydroxyl groups (**5**, **6** and **7**), ions corresponding to loss of H_2O , CO and ketene, are weak

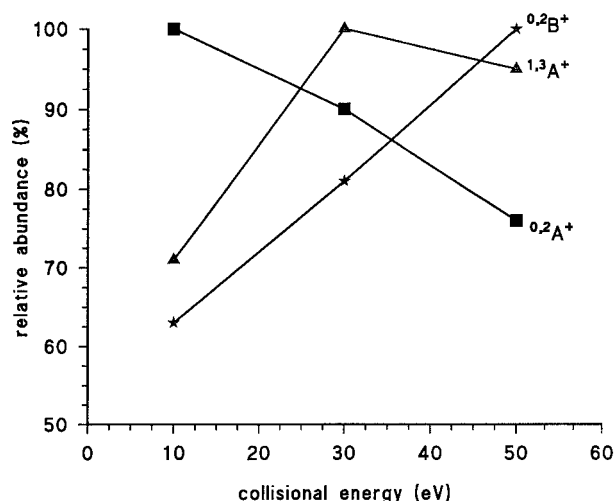


Figure 5. Relationship between relative abundance and collisional energy for the three major product ions $^{0.2}\text{A}^+$, $^{1.3}\text{A}^+$ and $^{0.2}\text{B}^+$ in the case of **5**.

(Table 2). The high mass region is dominated by the $[\text{M} - \text{CH}_3]^+$ ion (m/z 302); as found for methoxylated flavones, the loss of a CH_3 radical appears to be prevalent. The second most intense peak at m/z 285 corresponds to the loss of methanol and could be explained by an *ortho* elimination, which can be regarded as diagnostic of a vanillyl group. Further loss of CO yields the ion at m/z 257.

Direct characterization of flavone and flavonol aglycones in O-glycosides

A flavone mono-*O*-glycoside, luteolin-7-*O*-glucoside (**9**), and a flavonol diglycoside, kaempferol-7-*O*-neohesperidoside (**10**), have been examined in order to evaluate whether the aglycone part can be directly identified in *O*-glycosides without resorting to chemical hydrolysis. Compounds **9** and **10** both contain

different aglycones which have the same molecular weight. The positive-ion FAB mass spectra show a $[\text{M} + \text{H}]^+$ ion at m/z 449 and 595 for **9** and **10** respectively, and a protonated aglycone at m/z 287 (Y_0^+) (data not shown). Little structural information with regard to the aglycone part could be obtained from these spectra. The low-energy CID mass spectra obtained for the Y_0^+ ion are given in Fig. 6.

Although both Y_0^+ ions have the same m/z value (m/z 287), their product ion spectra under low-energy conditions are significantly different. For **9**, $^{1.3}\text{A}_0^+$, $^{1.3}\text{B}_0^+$, $^{0.2}\text{B}_0^+$, $^{0.4}\text{B}_0^+$ and $(^{0.4}\text{B}_0^+ - \text{H}_2\text{O})$ ions, at m/z 153, 135, 137, 179 and 161 respectively, are observed (Fig. 6), which are diagnostic of the flavone luteolin. For **10**, three main product ions $^{1.3}\text{A}_0^+$, $^{0.2}\text{A}_0^+$ and $^{0.2}\text{B}_0^+$, at m/z 153, 165 and 137 respectively, are found, which are typical of kaempferol, a flavonol.

CONCLUSIONS

The present data show that low-energy CID mass spectra are useful for characterization of, and differentiation between, flavone and flavonol aglycones. Under these conditions, simple and characteristic fragmentation patterns for each group are observed, which can be applied for fast identification and structural analysis of flavone and flavonol aglycones and their *O*-glycosides. A systematic ion nomenclature system is proposed in the present study, which is also suitable for designating the fragment ions produced using other soft ionization techniques, such as direct chemical ionization,¹⁹ atmospheric pressure chemical ionization,²⁰ thermospray,²¹ and electrospray.²²

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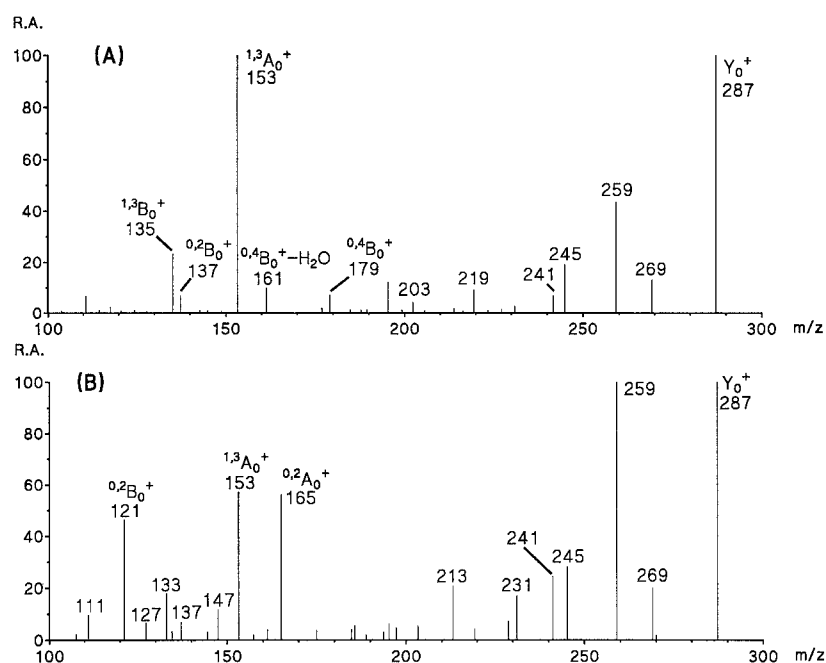


Figure 6. Low-energy CID spectrum obtained for the Y_0^+ ion (protonated aglycone) of (a) luteolin-7-*O*-glucoside and (b) kaempferol-7-*O*-neohesperidoside.

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